

Credit Fundamentals

Structure

- Introduction to credit
- Credit ratings – where macro and fundamentals come in
- Credit products and derivatives and how they are priced
- Idea generation

Introduction to Credit

Credit vs. Rates

- Credit – bet on the borrower to not default (and if they do, you cover enough for yourself)
- Rates – bet on the shape of yield curve (interest rates) of a country

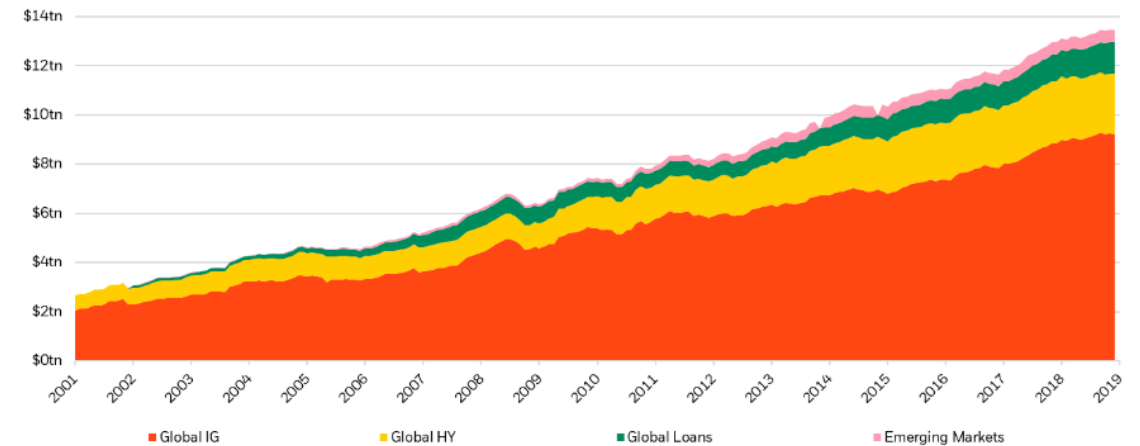
Categories

- Sovereign vs. Corporate
- Investment Grade vs. High Yield
 - Developed Markets vs. Emerging Markets
- Public (Bonds) vs. Private (Loans)
- Hard-currency vs. Local-currency

Why Invest in Credit?

Credit historically provides low or uncorrelated returns to other asset classes

	FI - Long	FI - Agg	Equity	Private Equity	Hedge Funds	Real Estate	Cash
FI - Long							
FI - Agg	0.93						
Equity	0.07	0.10					
Private Equity	0.02	0.04	0.82				
Hedge Funds	0.04	0.06	0.82	0.70			
Real Estate	0.26	0.29	0.74	0.49	0.52		
Cash	-0.07	-0.01	-0.22	-0.27	-0.27	-0.10	
Credit	0.18	0.22	0.61	0.44	0.55	0.53	-0.18

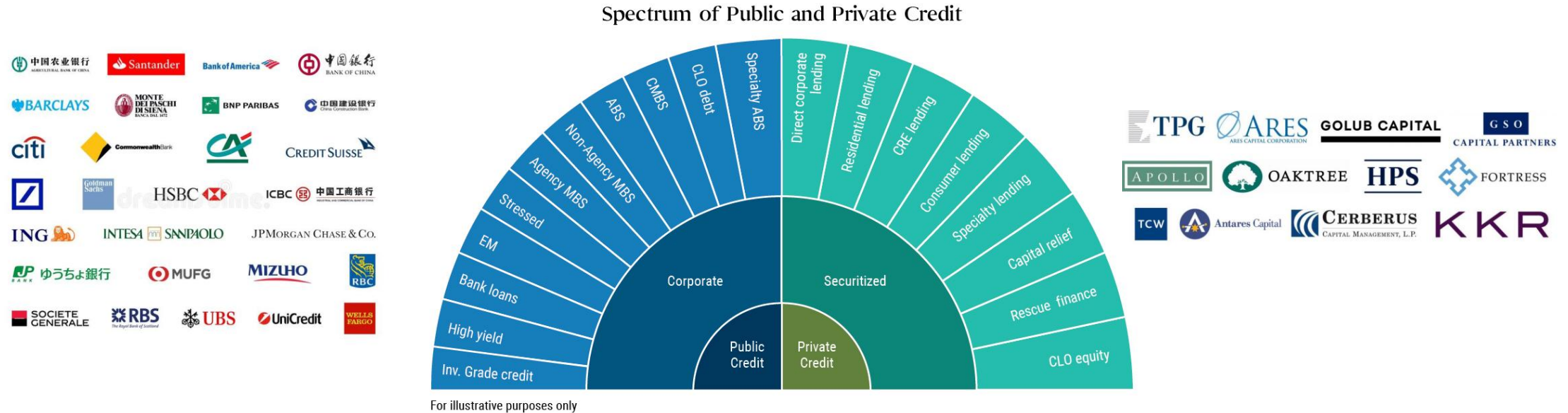


- Good diversification benefits
- Stable stream of income (yield)
- Access to borrower-specific features

Credit as a Career

- Sales & Trading – secondary market
- “Investment Banking” – primary market
 - Leveraged Finance
 - DCM
- Buy Side
 - Private Credit
 - Liquid Credit
 - Distressed Debt / Special Situations

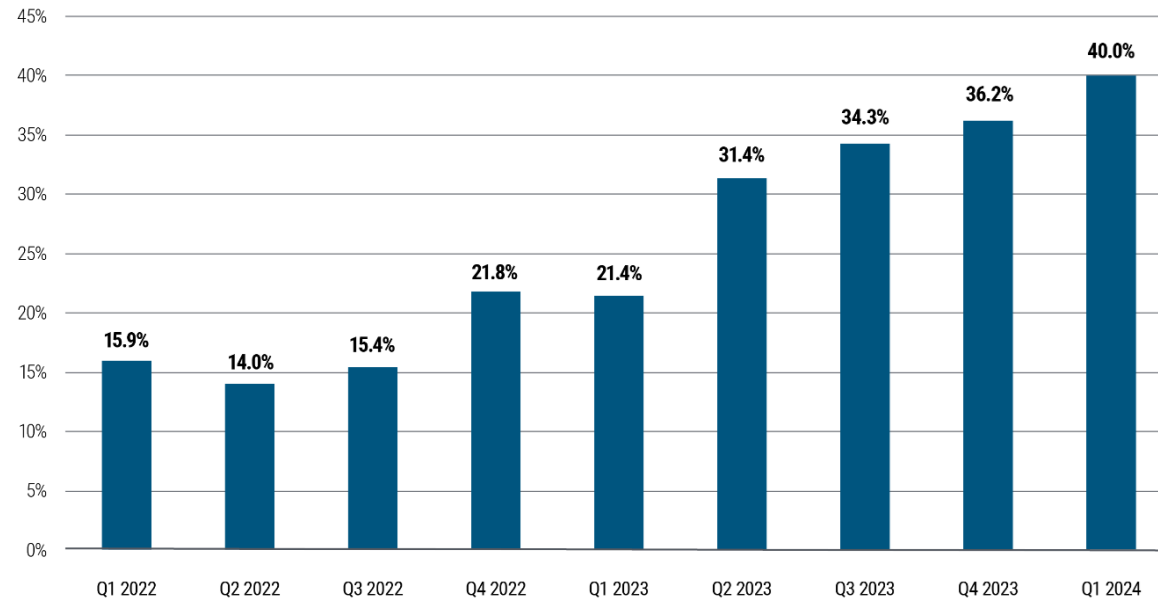
Public vs Private



- Public market debt mostly issued by “sell-side” banks, traded on public exchanges and so are liquid, subject to regulatory and disclosure requirements
- Private market debt is provided by “buy-side” firms – products are not traded and illiquid, unrated

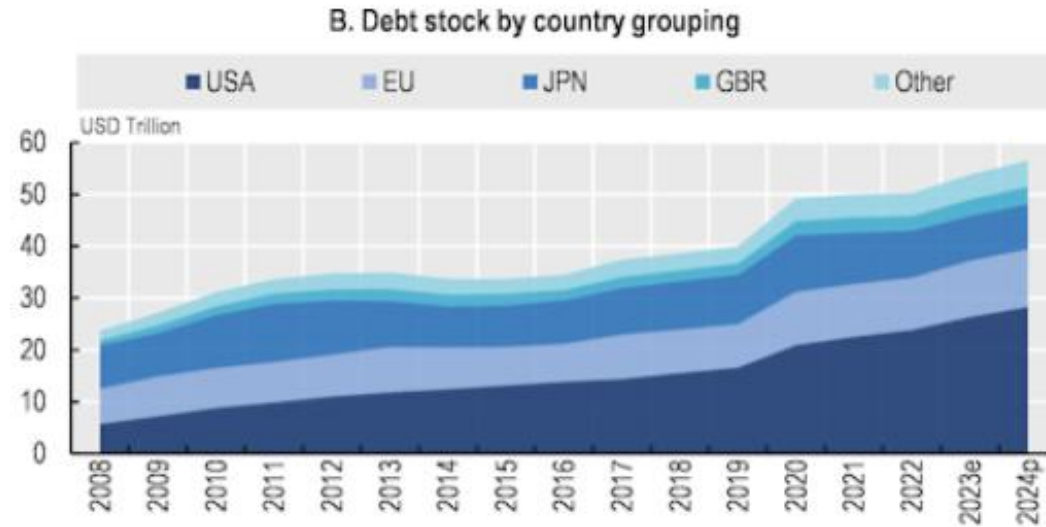
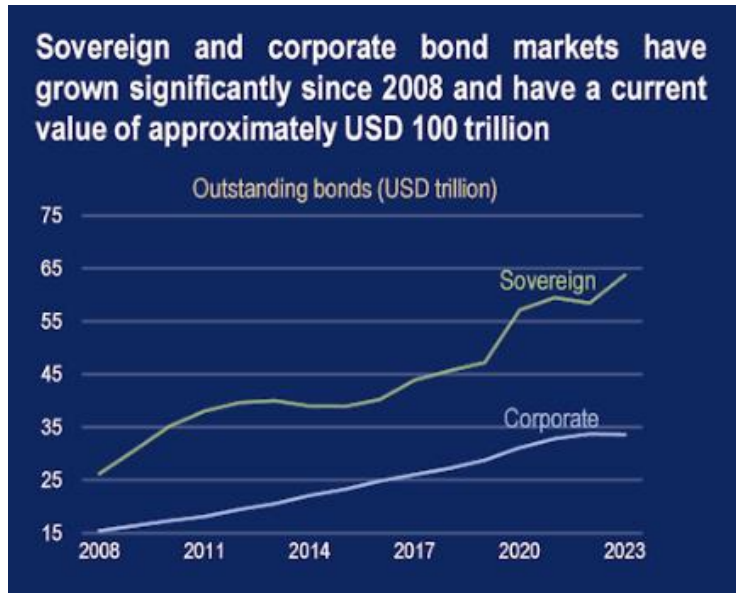
Private credit market

Percentage of private credit borrowers with fixed charge coverage ratios under 1x



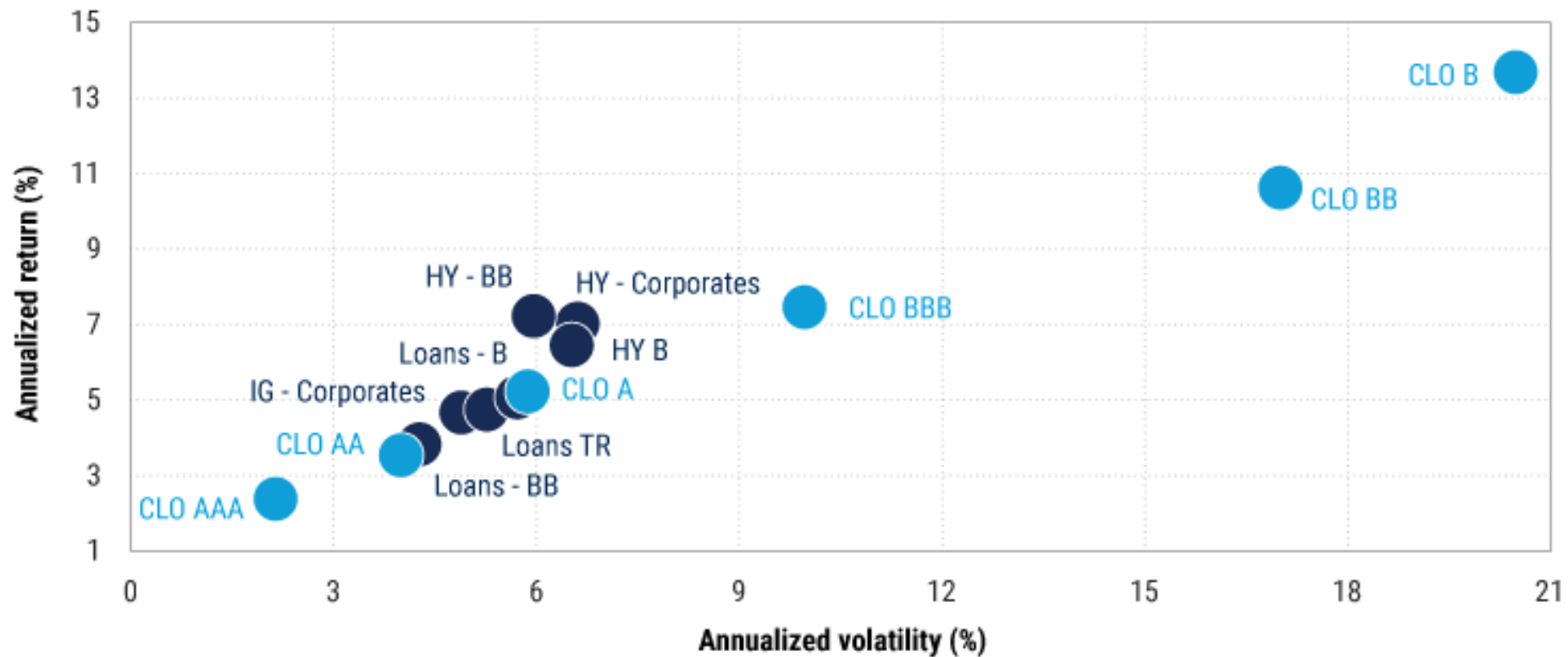
- Banks have retreated from traditional lending post-Dodd Frank
- Boom over the past few years and likely to continue
- Heavily concentrated in sectors such as tech and health services
- Typically issued by smaller, leveraged companies financed with floating-rate debt

Sovereign credit



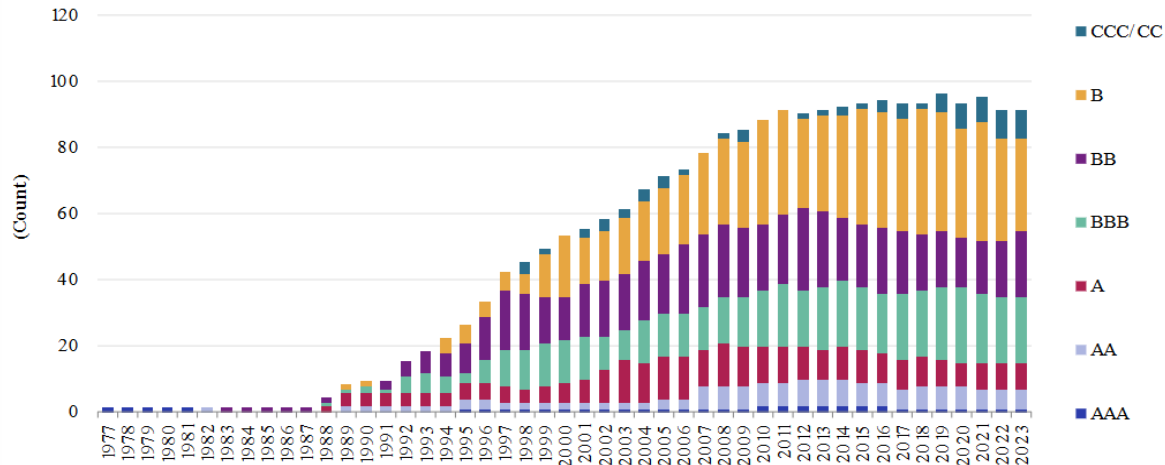
- Governments (developed or emerging) raise money in their own currency (local currency) or another currency (hard currency, usually USD and more common for EM)
- Governments can borrow domestically, but due to limited funds available may also market their debt internationally or seek out arrangements with other govts/international organisations

Risk-return scatterplot for credit investments

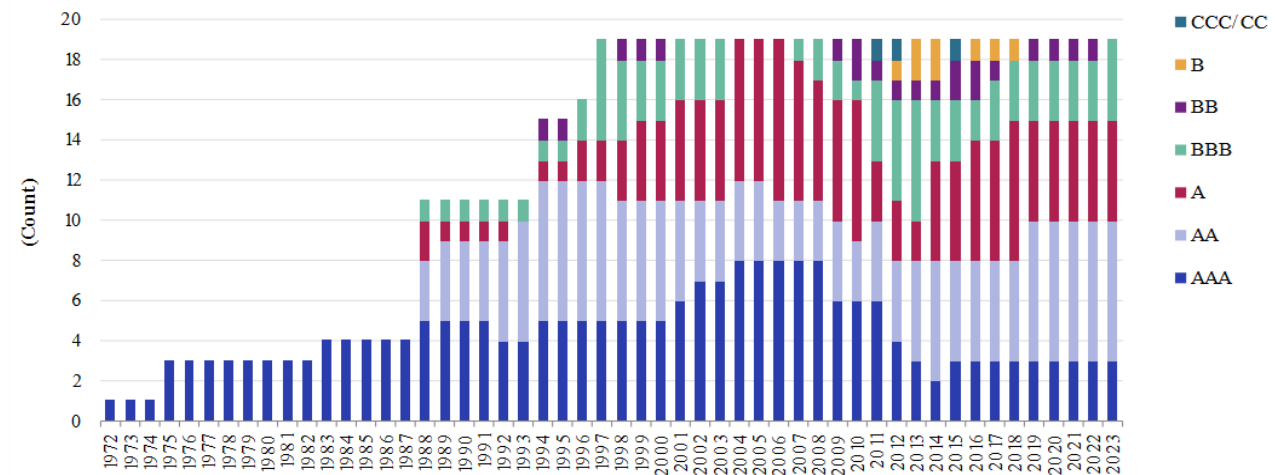


Developed vs. emerging markets

Distribution of emerging and frontier market sovereign ratings



Distribution of eurozone sovereign ratings



- Developed markets: highly liquid government bond markets, stable economic conditions, and low credit risk
 - USTs, JGBs, Eurozone debt (but country credit risk can vary; Germany as benchmark)
- Emerging markets: higher yields due to greater credit and currency risks, number of investable EM countries has more than doubled in past 20 years.
 - LatAm (Brazil, Mexico, Argentina), Eastern Europe, some ASEAN (Indonesia, Malaysia)
 - More financially advanced countries can target domestic investors, others have less developed domestic capital market so issue bonds externally in hard currency

Credit Analysis

5Cs of basic credit analysis

The 5 C's of Credit

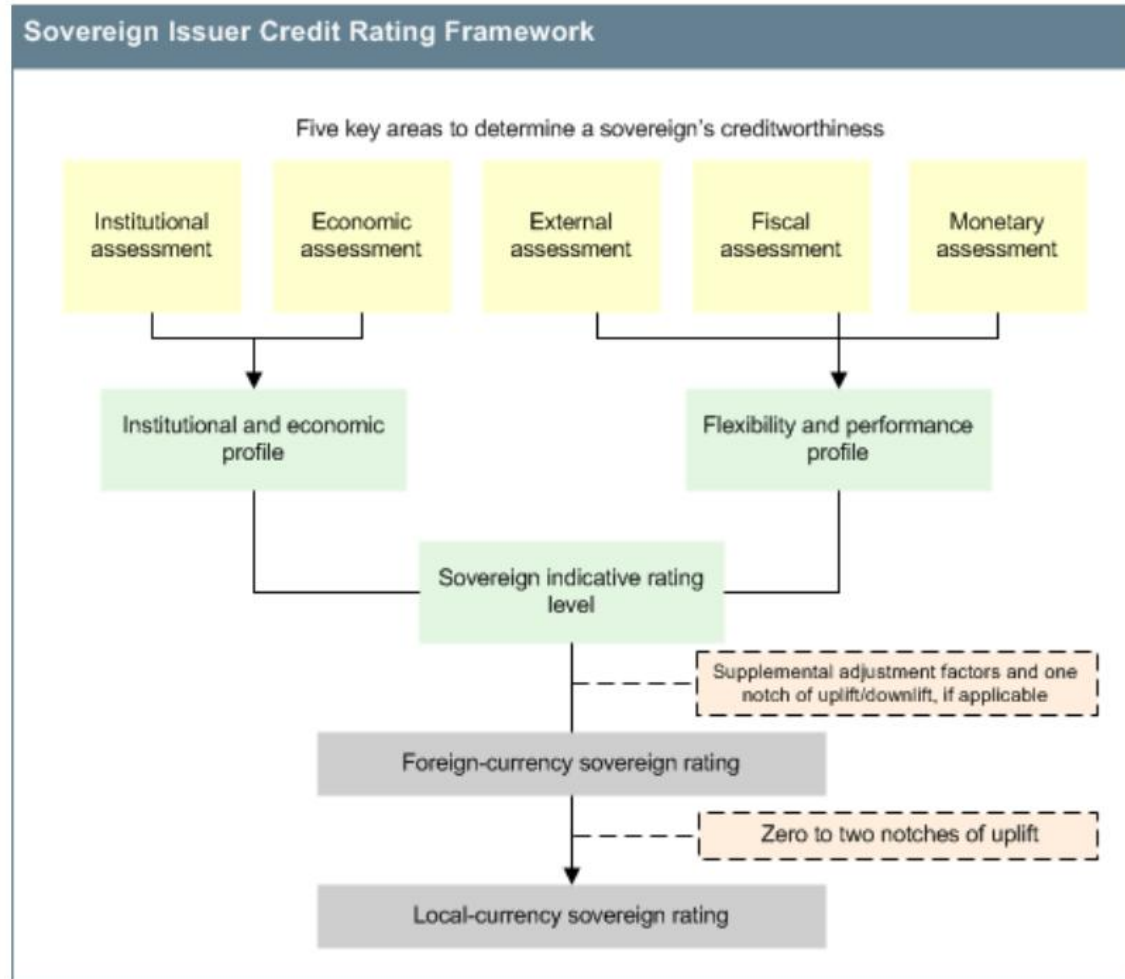


Credit Ratings

	MOODY'S	S&P Global	FitchRatings
	Moody's	S&P	Fitch
Investment Grade	Aaa	AAA	AAA
	Aa1	AA+	AA+
	Aa2	AA	AA
	Aa3	AA-	AA-
	A1	A+	A+
	A2	A	A
	A3	A-	A-
	Baa1	BBB+	BBB+
	Baa2	BBB	BBB
	Baa3	BBB-	BBB-
Non-Investment Grade <i>High Yield "Junk"</i>	Ba1	BB+	BB+
	Ba2	BB	BB
	Ba3	BB-	BB-
	B1	B+	B+
	B2	B	B
	B3	B-	B-
Default	Caa	CCC	CCC
	Ca	CC	CC
	C	C	C
	C	D	D

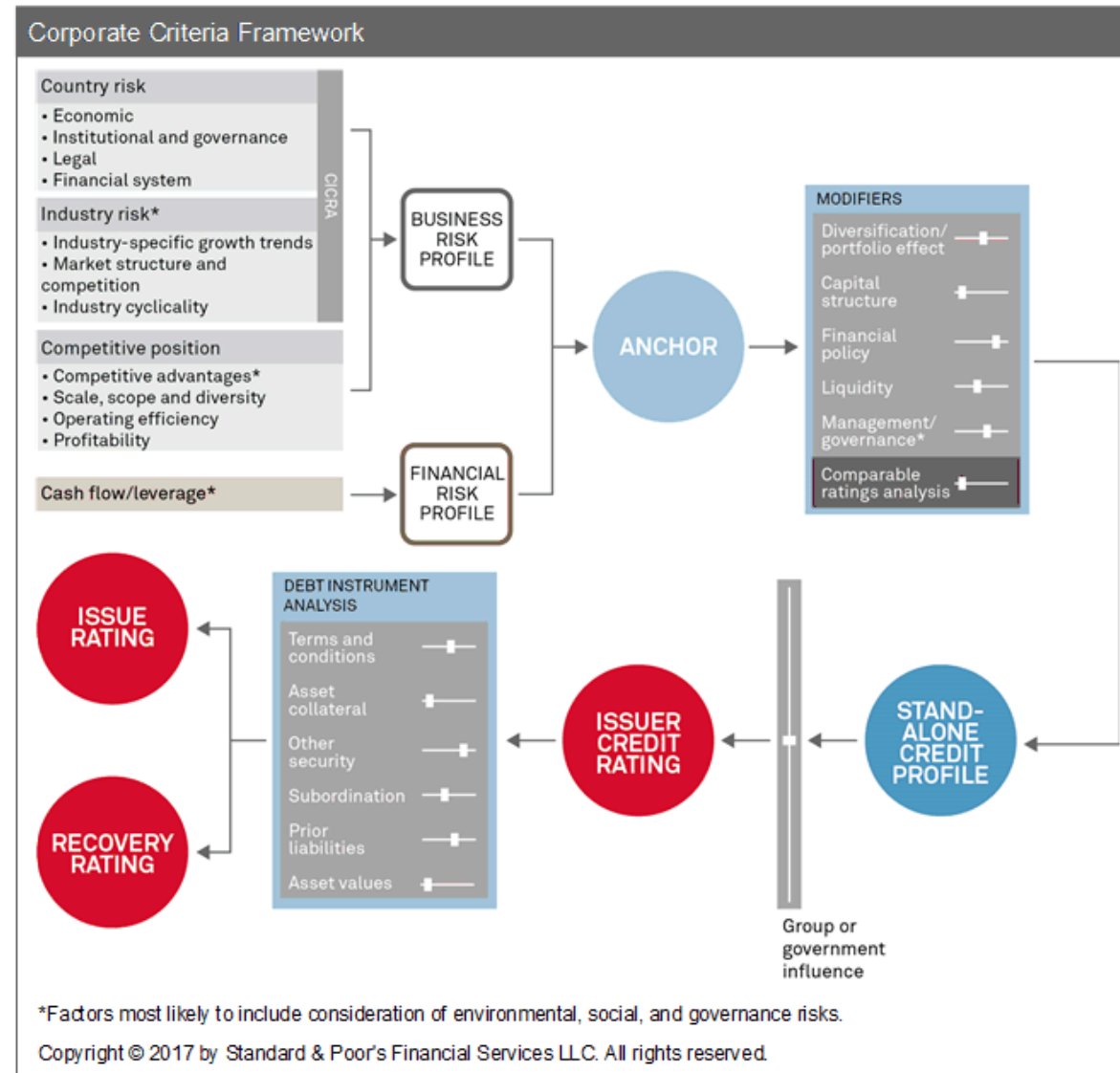
- **Risk assessment** for investors seeking portfolio strategy and diversification
- Higher credit ratings lead to lower interest rates and thus **lower borrowing costs**
- **Eligibility** for inclusion into different bond indices

Credit Rating Methodology – S&P



- Country of incorporation and currency that bond is raised in is very important
- Sovereign rating determines risk-free rate for the country – becoming the base of cost of funding

Credit Rating Methodology – S&P



- Start with macro – country and industry that company operates in
- Continue with micro – company ratios that would affect repayment
- Finish with individual credit analysis – structure and terms of credit product

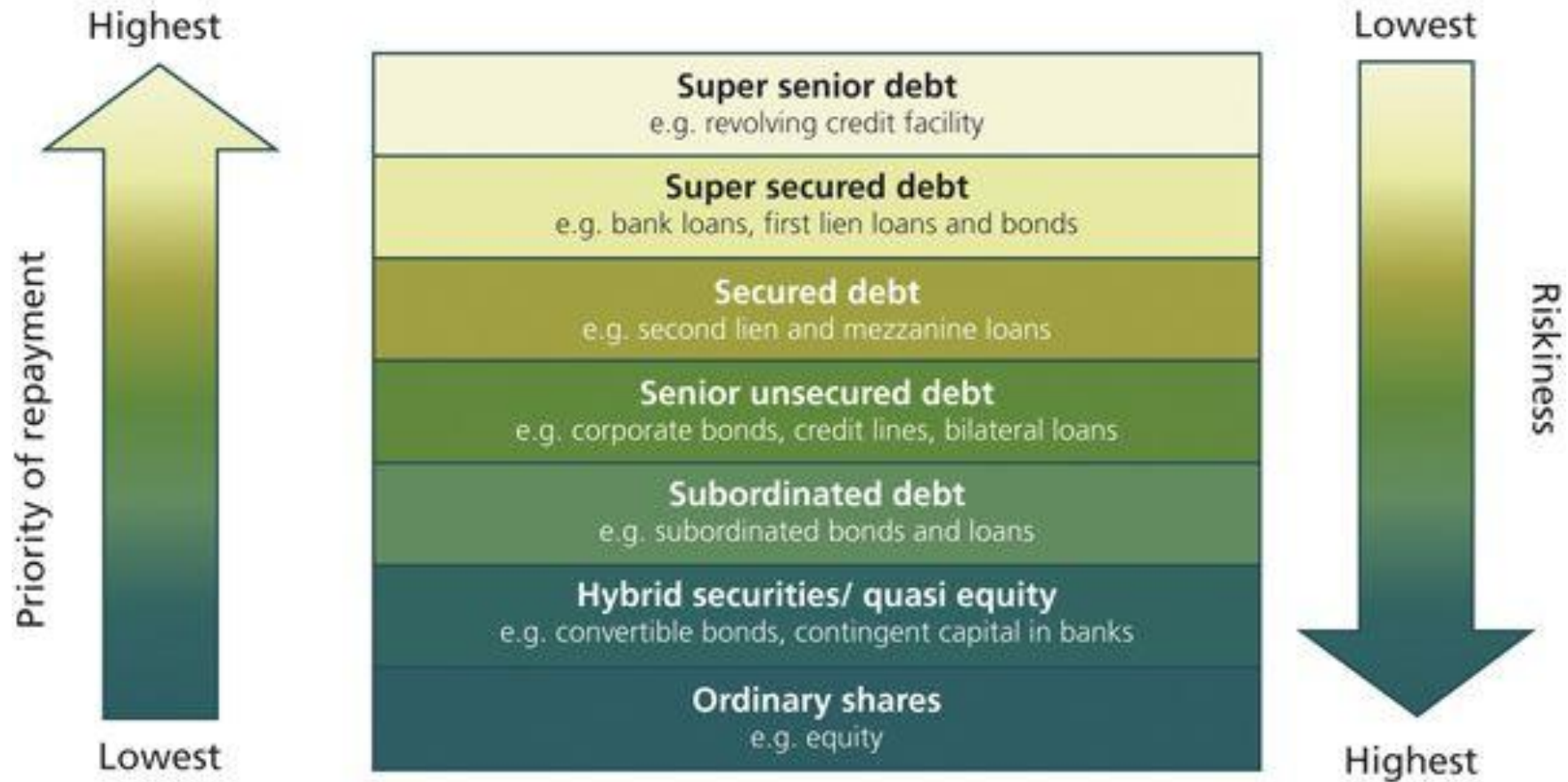
Basic Jargon

1. Covenants: Set of rules that the issuer must follow

Split into Financial and Restrictive Covenants

2. EBITDA: Earnings before Interest, Taxation, Depreciation and Amortisation
3. Leverage Ratio: $\text{Debt} / \text{EBITDA}$
4. Interest Coverage Ratio: $\text{EBIT} / \text{Annual Interest Expense}$
5. Debt Service Coverage Ratio: $\text{EBITDA} / \text{Interest} + \text{Principal Repayments}$

Capital Structure



- Main difference between bonds and loans is that bonds are traded publicly through an exchange, while loans are not traded publicly.
- Bonds are typically unsecured while loans are generally secured by collateral

Bond terms & structure

- Floating rate: e.g. LIBOR + 200 or fixed rate
- Call provision: Bond can be called by issuer – usually to refinance at lower rate
- Convertibility: after a certain date or upon fulfilling a condition, into certain number of shares
- Liquidation preference: Senior > Junior > Hybrid (Convertibles)
- Credit enhancement: guarantee, letter of credit, collateral

ISSUER	Commonwealth of Australia
INSTRUMENT	Treasury Bonds
CURRENCY	Australian dollars
SERIES NUMBER	TB140
AUSTRACLEAR ISIN	AU000XCLWAG2
ASX CDI ISIN	AU000GSBG333
ASX CODE FOR EXCHANGE-TRADED TREASURY BOND	GSBG33
FIRST ISSUE DATE	19 November 2013
MATURITY DATE	21 April 2033
COUPON	4.50% per annum paid semi-annually in arrears on the Face Value of the bonds.
REDEMPTION	Par
COUPON PAYMENT DATES	21 April and 21 October in each year commencing on 21 April 2014, to and including the Maturity Date
DAY COUNT	Actual/Actual
BUSINESS DAYS	Sydney

Basic Bond Valuation

Present Value of Discounted Cash Flows

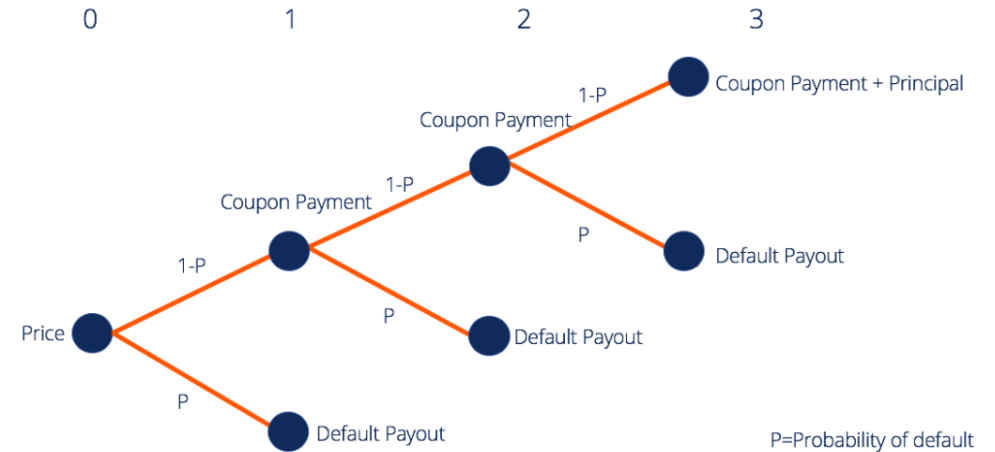
$$PV = \frac{CF_1}{(1+r)^1} + \frac{CF_2}{(1+r)^2} + \frac{CF_3}{(1+r)^3} \dots \frac{CF_n}{(1+r)^n}$$

CF equals cash flow for a period,

r equals the discount rate, and

n equals the number of periods.

Period (Years)



- Simple DCF, with simulated future default rates
- Discounted value of future expected payments (interest and principal) received

Understanding Pricing/Yields

Trader 1: Got any Google 4 ¼ 27's?

Trader 2: Offering 50 at 98 ¾

Average Price = $(100 + 98.75)/2 = 99.375$

Average Yield = $4.25 + (100 - 98.75)/2 = 4.875\%$

YTM = $4.875\% / 99.375 = 4.91\%$

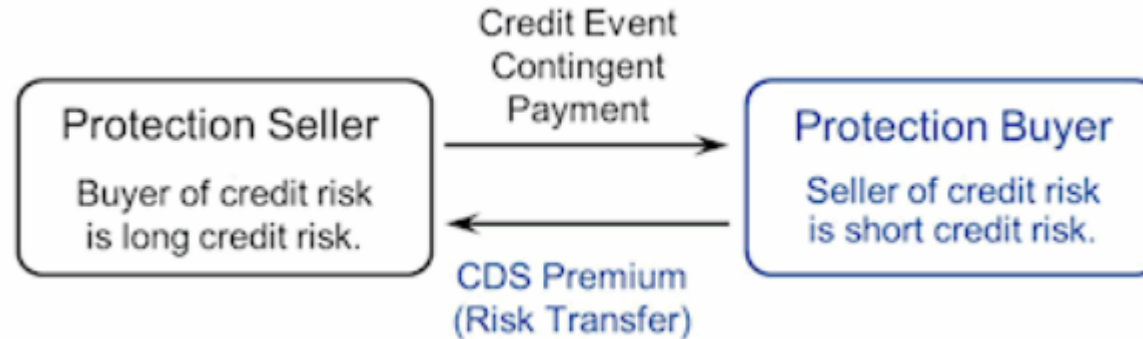
Actual YTM = 4.92%

Credit Products

Credit products

- Vanilla bond e.g. Bank of America 5y note
 - Traded in lot sizes of USD100k, USD250k, USD1m etc.
 - Purchased by HNW individuals, asset managers, insurance companies etc.
- Securitized products
 - ABS – asset backed securities - loans, leases, receivables
 - MBS – mortgage backed securities
- Convertible bonds
 - Can be converted into equity

Credit derivatives – Credit default swaps

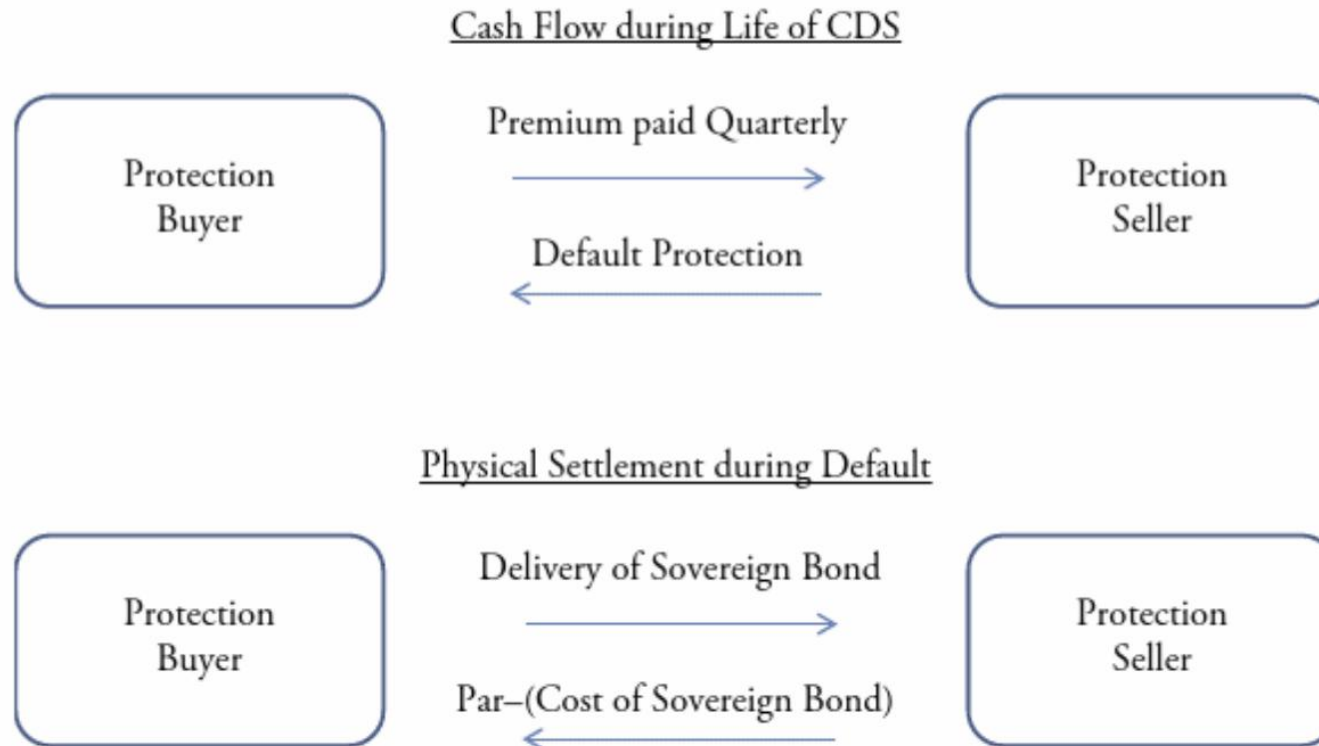


$$CDS \text{ Spread (Premium)} = \text{Default Probability} * (1 - \text{Recovery Rate})$$

- Premium paid is equal to expected loss from default – theoretically assuming risk neutrality – “credit insurance”
 - By speculating on loss on default, can generate alpha from buying/selling CDS
 - Example triggers: ratings downgrade, event of default, cross-default
 - Only liquid for very large corporates e.g. Apple, otherwise OTC
- Industry practice to pay 100bps per year as premium and present value of the actual premium upfront

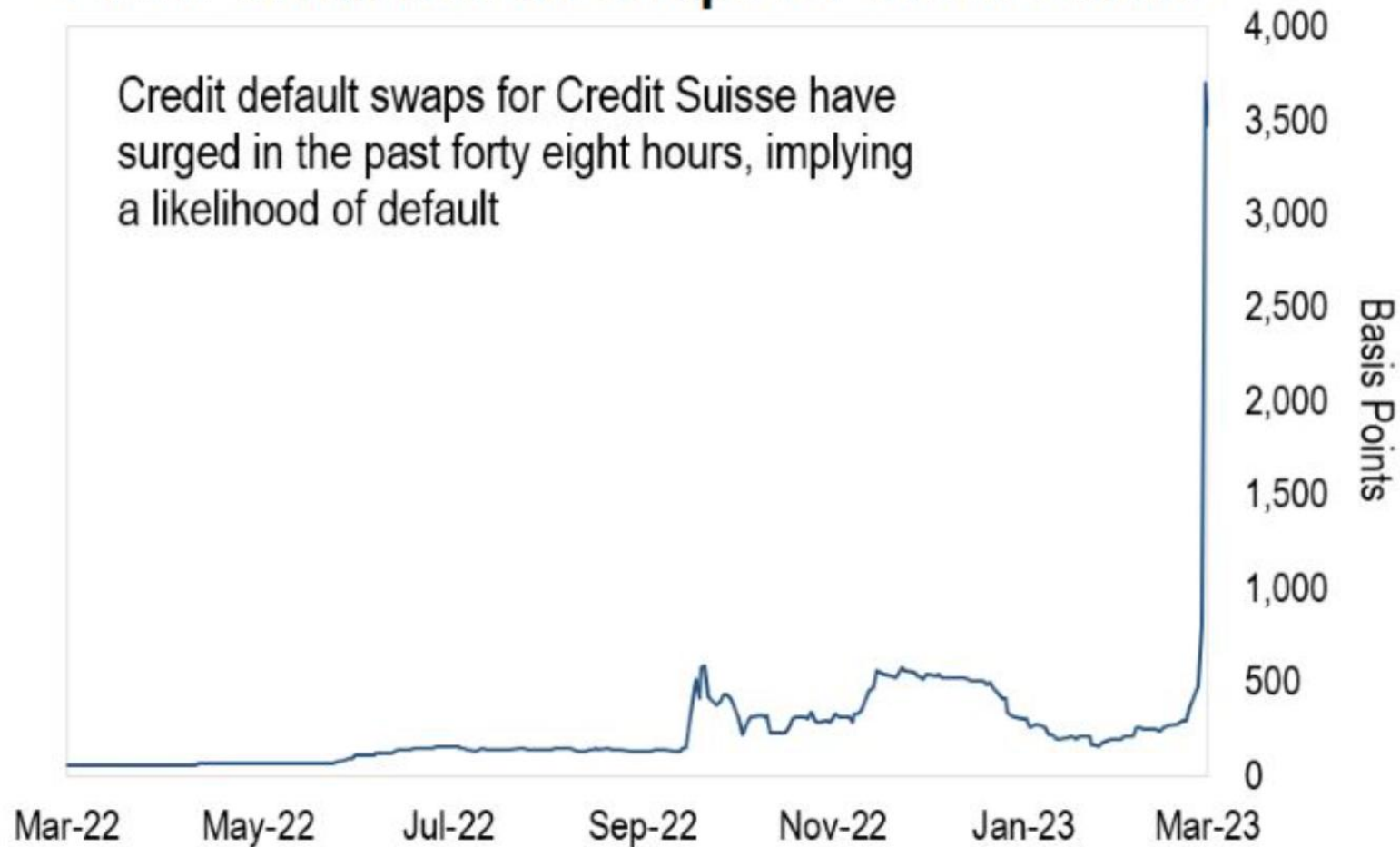
Credit derivatives – Credit default swaps

- Credit risk (counterparty risk)- risk that the company defaults (failure to repay or meet contractual obligations)



CDS for CS 2022-2023

1-Year Credit Default Swaps for Credit Suisse

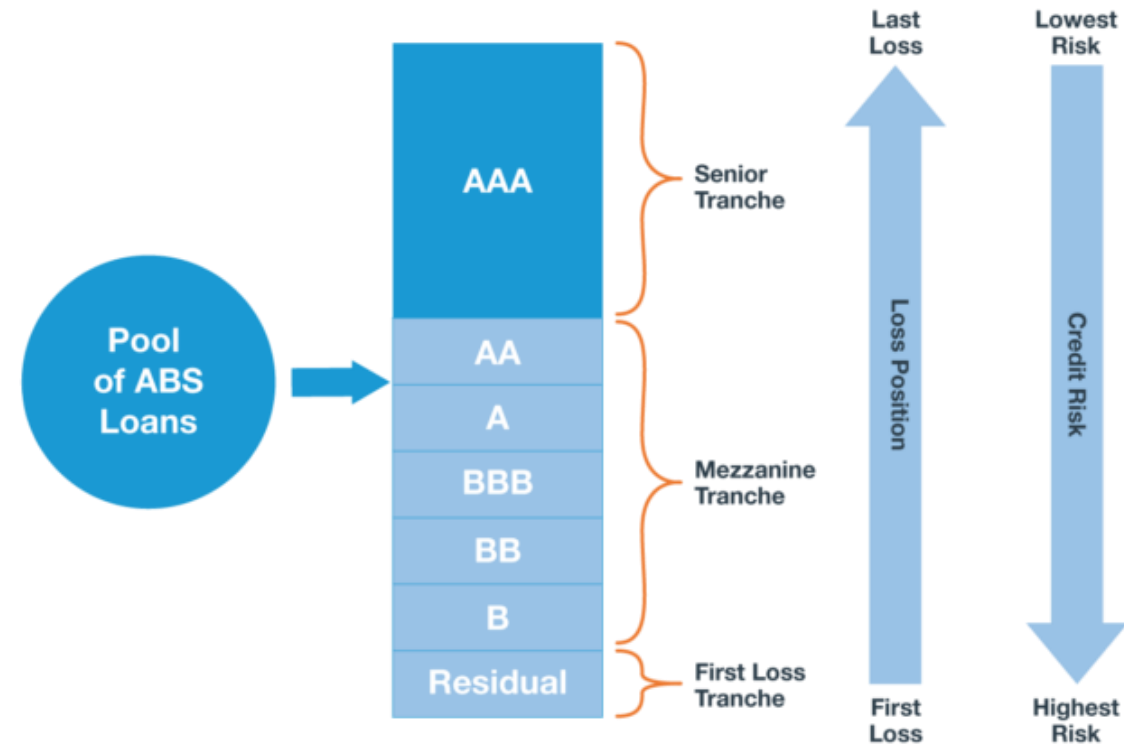


Asset-Backed Securities ABS

Common types:

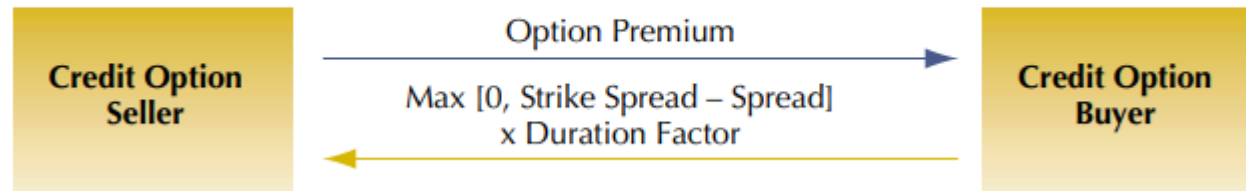
1. Collateralized debt obligation (issued by a special purpose vehicle)
 - a. Loans, bonds, cash-market debt instruments, mortgages
2. Home equity loans (for those who did not qualify for a mortgage)
3. Auto loan
4. Student loan

Priced by underwriters and administered by servicers

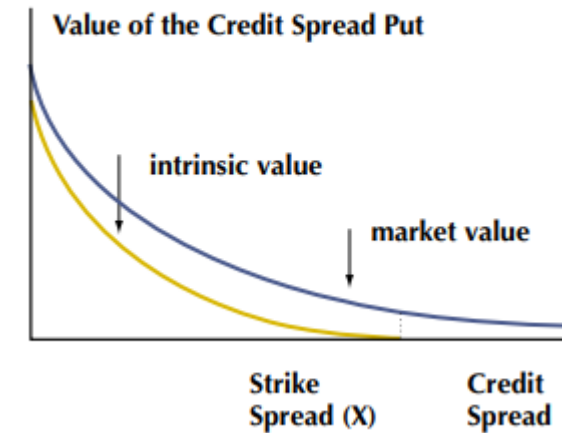


Credit derivatives – credit spread options

A Credit Spread Call Option



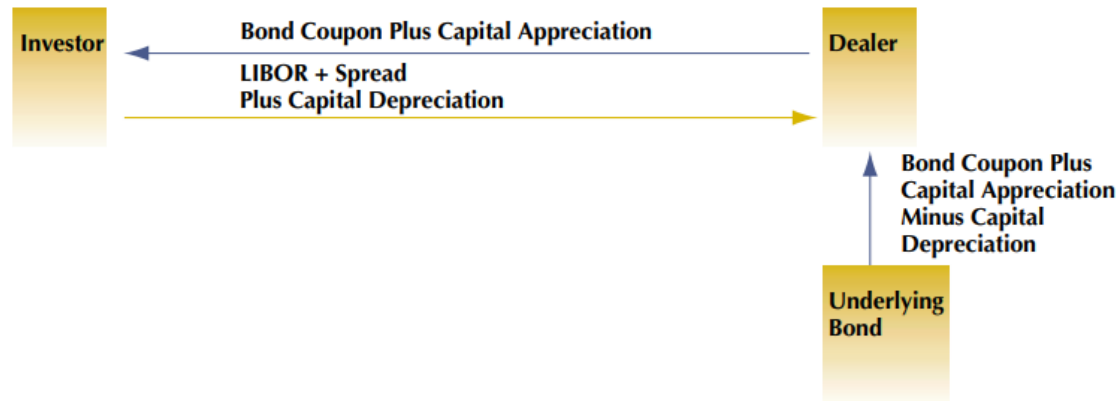
The Value of a Credit Spread Call on a Nonredeemable Bond



- Similar to equity puts & calls but now the strike is a spread level
 - Usually, spread is between company's debt and some benchmark level (e.g. LIBOR)
- Credit spread puts increase in value as the yield spread on a bond increases
- Credit spread calls increase in value as the yield spread on a bond decreases

Credit derivatives – Swaps

Total return swap



Asset swap



Result: The investor has purchased a fixed-rate asset and converted it to a floating-rate asset.

- Total return swaps utilise the bank's balance sheet – essentially provides leverage as investor can obtain the asset's return without owning it
- Asset swaps allow investors to exchange the fixed payment of their assets into a floating one – allowing them to speculate

Fixed Income ETFs

- Credit index ETFs can be used to gain exposure to various diversified fixed income portfolios (it replicates an underlying bond index, and it's more liquid)
- List of common fixed income ETFs

Symbol	Description
SHY <Equity>	iShares 1–3 Year Treasury Bonds
IEF <Equity>	iShares 7–10 Year Treasury Bonds
TLT <Equity>	iShares 20+ Year Treasury Bonds
TBT <Equity>	ProShares UltraShort 20+ Year Treasuries
AGG <Equity>	iShares Core Total Aggregate U.S. Bonds
LQD <Equity>	iShares iBoxx \$ Investment Grade Corporate Bonds
TIP <Equity>	iShares TIPS Bond
MBB <Equity>	iShares MBS
MUB <Equity>	iShares National AMT-Free Municipal Bonds
LQD <Equity>	iShares iBoxx \$ Investment Grade Corporate Bonds
PCY <Equity>	PowerShares Emerging Markets Sovereign Debt
HYG <Equity>	iShares iBoxx \$ High Yield Corporate Bonds
JNK <Equity>	SPDR Barclays High Yield Bonds
BKLN <Equity>	PowerShares Senior Loan Portfolio

Idea Generation

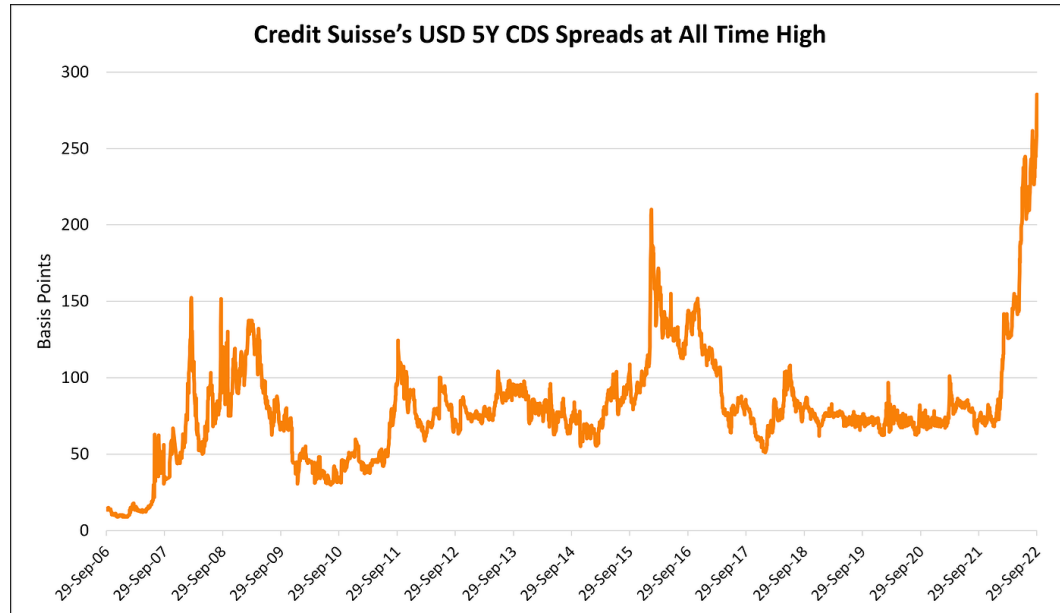
Type 1: Single Company Fundamentals

- Trade ideas based on change in company's fundamentals which may affect future repayment/default probability
- Bond prices are direct indicators of credit quality, barring any changes to the underlying base rate. E.g. If a bond is trading at 70, the company is probably distressed.
- If you believe credit conditions will improve, you long
- If you believe otherwise, you short

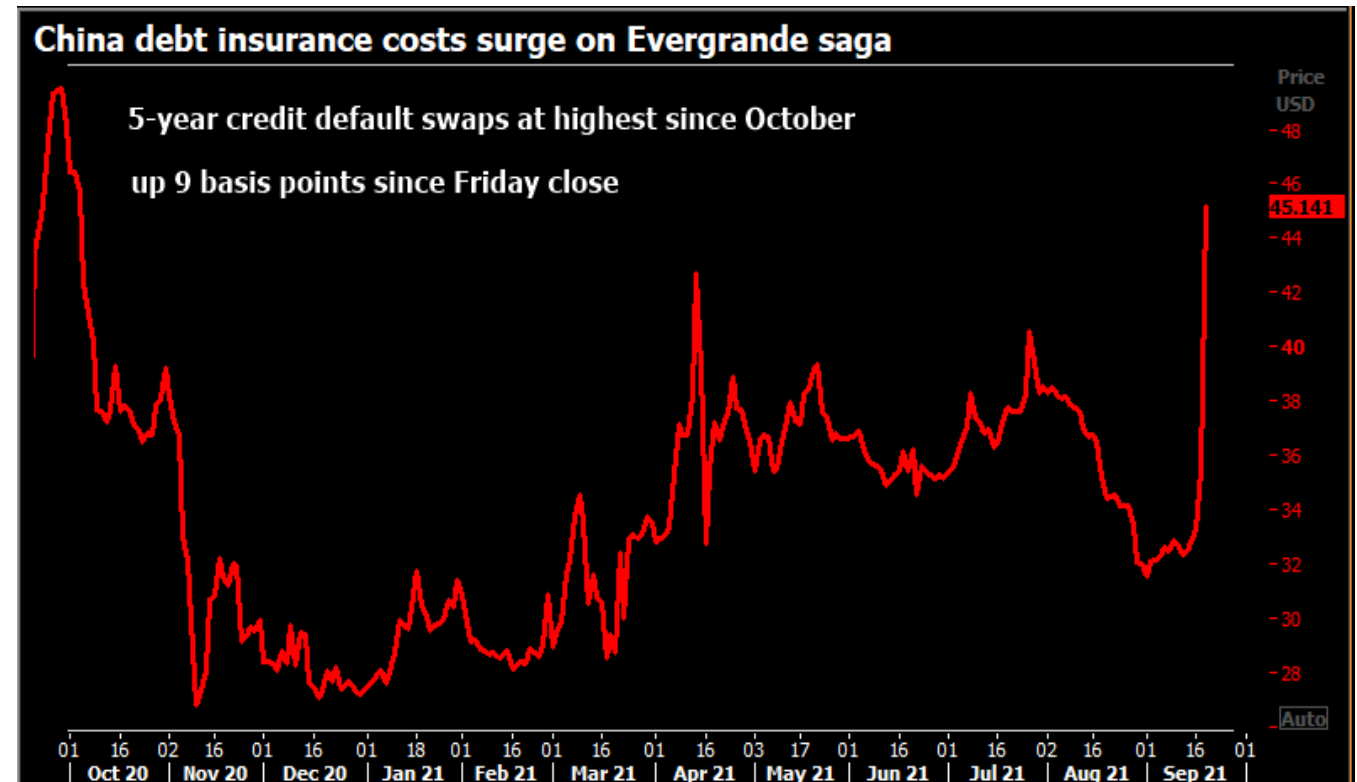
Type 1: Single Company Fundamentals

- Longing a bond is very straightforward, you just collect the yield and wait for price appreciation
- Shorting is much more complicated, because it requires you selling the bond and then buying it back at a lower price
- Unlike shorting a stock, shorting a bond requires you to pay the coupon during the duration of the whole trade. This gets expensive quickly, especially amongst high-yield issuers.
- Hence, the more efficient way is to purchase a CDS on that specific credit instead.

Type 1: Single Company Fundamentals



Source: Bloomberg, BondValue



Type 2: Mispricings

- Pricing of public bonds and credit derivatives are often based on credit ratings from reputable firms
 - Banks have their own internal ratings system to manage their portfolio of debtors and to make lending decisions
 - Private credit firms have to make their own assessments of firm default probability to price private credit
- Generating alpha by locating mispricings

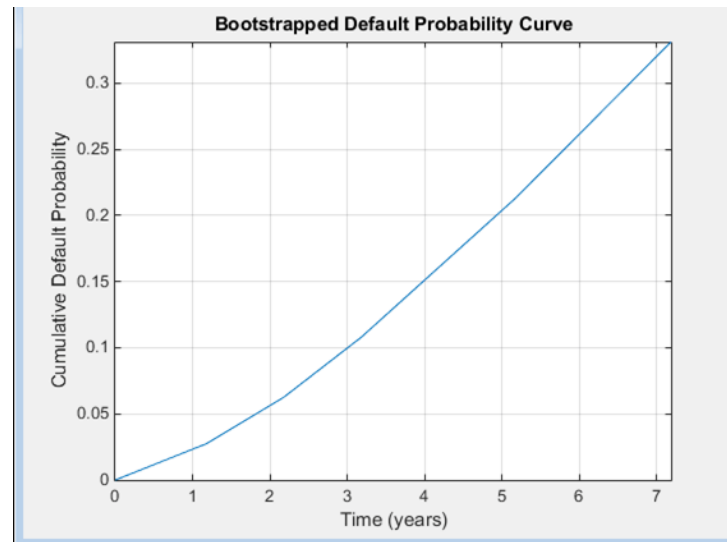
Global corporate average cumulative default rates by rating modifier (1981 - 2017)

Credit rating	Time horizon									
	Time horizon									
(%)	1	2	3	4	5	6	7	8	9	10
AAA	0.00	0.03	0.13	0.24	0.35	0.46	0.51	0.60	0.65	0.71
AA+	0.00	0.05	0.05	0.10	0.16	0.21	0.27	0.33	0.39	0.45
AA	0.02	0.03	0.08	0.22	0.36	0.48	0.61	0.72	0.81	0.91
AA-	0.03	0.09	0.18	0.25	0.33	0.45	0.52	0.57	0.63	0.69
A+	0.05	0.09	0.20	0.34	0.45	0.55	0.66	0.79	0.93	1.08
A	0.06	0.15	0.24	0.36	0.49	0.68	0.86	1.03	1.23	1.47
A-	0.07	0.17	0.28	0.40	0.57	0.74	0.98	1.16	1.30	1.42
BBB+	0.11	0.31	0.53	0.77	1.03	1.32	1.54	1.78	2.04	2.30
BBB	0.17	0.43	0.68	1.05	1.42	1.80	2.15	2.49	2.85	3.23
BBB-	0.25	0.77	1.39	2.11	2.84	3.50	4.09	4.65	5.11	5.53
BB+	0.34	1.11	2.02	2.94	3.86	4.74	5.50	6.05	6.70	7.33
BB	0.56	1.71	3.38	4.94	6.52	7.77	8.89	9.85	10.75	11.53
BB-	1.00	3.13	5.37	7.66	9.66	11.62	13.24	14.80	16.04	17.12
B+	2.08	5.71	9.23	12.21	14.53	16.33	17.98	19.43	20.77	21.97
B	3.60	8.29	12.29	15.46	17.89	20.15	21.66	22.76	23.77	24.81
B-	7.15	14.28	19.62	23.37	26.18	28.31	29.99	31.13	31.84	32.40
CCC/C	26.82	36.03	41.03	43.97	46.22	47.13	48.33	49.23	50.08	50.71
Investment grade	0.10	0.26	0.45	0.68	0.92	1.17	1.40	1.61	1.82	2.03
Speculative grade	3.75	7.31	10.39	12.90	14.95	16.64	18.05	19.23	20.27	21.21
All rated	1.50	2.95	4.22	5.29	6.18	6.94	7.57	8.12	8.60	9.05

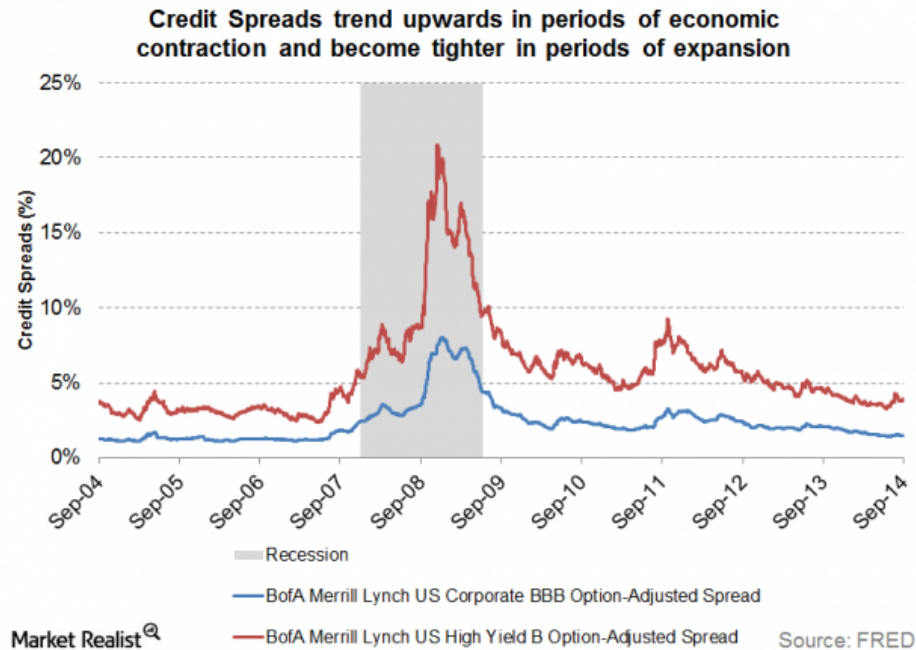
Source: Standard and Poor's Global Research

Type 2: Mispricings

- Finding the market default rates of any firm
 - Market CDS premium
 - How much their bonds are trading at
 - Recent loan/financing rate
 - Recovery rate – generally set at 40%



Idea generation – macro driven



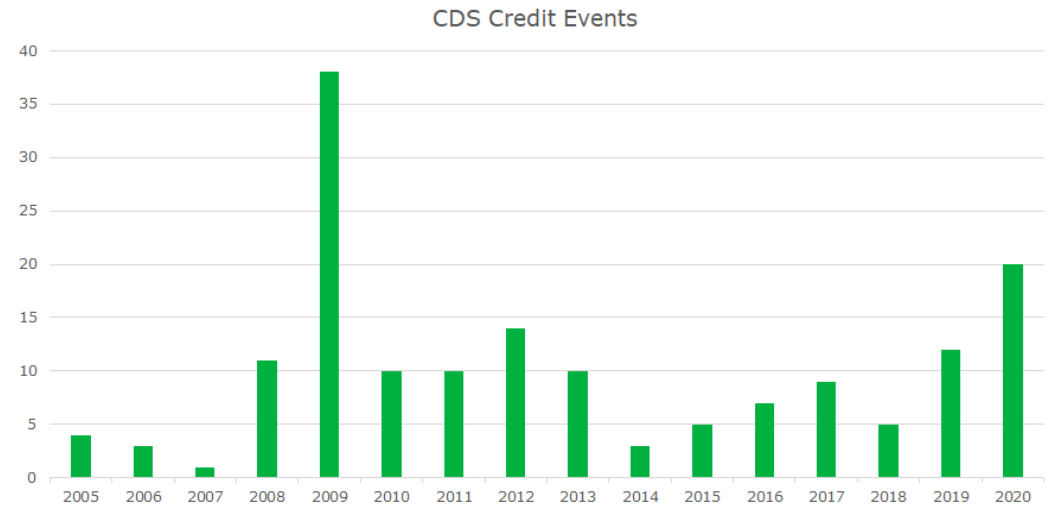
- Predicting recessions, political events, business cycles where industries perform poorer – credit is a cyclical market where spreads will reflect macro outlook
- Market confidence in credit will be reflected in credit spreads

Idea generation – steepeners/flatteners



- Steepener: Short short term CDS and Long long term CDS
- A play on general market recovery/recession or mispricing of certain ends of the curve, can look at business cycles
- Generally only 5y CDS is liquid, sometimes 1y and 10y as well

Case study – 08' financial crisis



- CDS trades shown in The Big Short (2015) and The Greatest Trade Ever (2009)
- Both cases investors made bets against the US housing market and that people would default on their mortgages

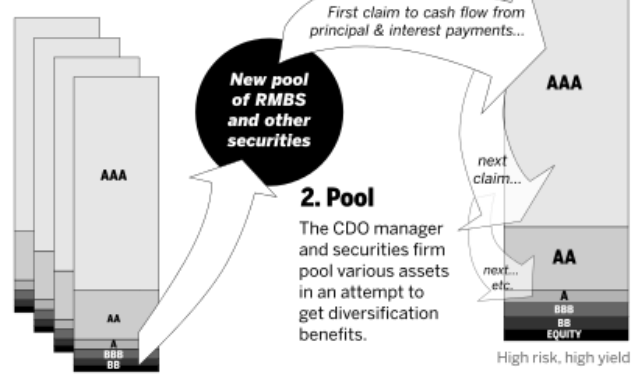
Case study – 08' financial crisis

Collateralized Debt Obligations

Collateralized debt obligations (CDOs) are structured financial instruments that purchase and pool financial assets such as the riskier tranches of various mortgage-backed securities.

1. Purchase

The CDO manager and securities firm select and purchase assets, such as some of the lower-rated tranches of mortgage-backed securities.



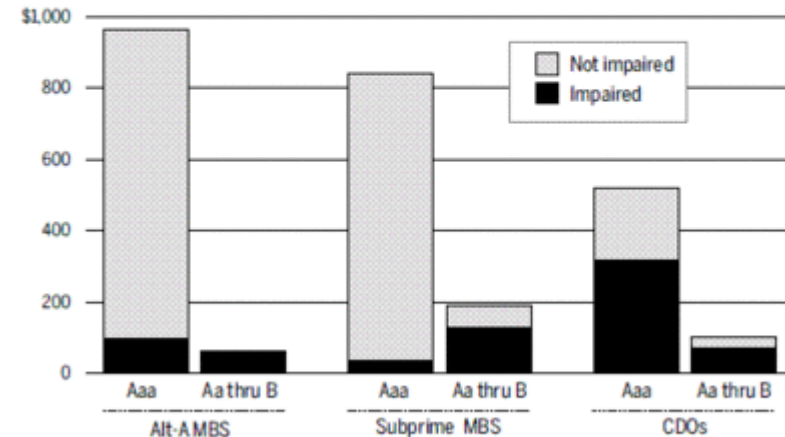
3. CDO tranches

Similar to mortgage-backed securities, the CDO issues securities in tranches that vary based on their place in the cash flow waterfall.

Impaired Securities

Impairment of 2005-2007 vintage mortgage-backed securities (MBS) and CDOs as of year-end 2009, by initial rating. A security is impaired when it is downgraded to C or Ca, or when it suffers a principal loss.

IN BILLIONS OF DOLLARS



SOURCE: Moody's Investors Service, "Special Comment: Default & Loss Rates of Structured Finance Securities: 1993-2009"; Moody's SFDRS.

- CDS on subprime mortgage-backed securities
- Came at a low price as these were highly rated securities by ratings firms – poor quality assets, but they were highly diversified